

AVIAD ARANIAS

Machine Learning Engineer | World Models, Real-Time Video ML & GPU Infrastructure at Scale

Brooklyn, NY • 347-863-2860 • aviadarn@gmail.com • linkedin.com/in/aviad-a-b0606682 • github.com/aviadarn

SUMMARY

ML engineer who fine-tunes action-conditioned world models at home and spent seven years running real-time video ML in production at Nielsen Sports — 10K+ RPS of GPU inference on a 30-node, 100+ GPU Kubernetes cluster, serving a live broadcast audience 24/7. Fine-tuned iVideoGPT, an autoregressive transformer world model, on the Bridge V2 robot-manipulation dataset with a fully self-built data-preparation, training, and evaluation pipeline — beating the pretrained baseline on every rollout metric. Learned the hard way — on a robot I built with my kid — that dataset distribution, not architecture, is usually what makes a policy work in the physical world. Hands-on across the entire stack: dataset curation and labeling, PyTorch training on Ray across 100M+ image datasets, TensorRT FP16/INT8 quantization, a custom C++/CUDA inference server, Kafka streaming ingestion of continuous multi-camera video, and edge deployment on a 10-watt Jetson. M.Sc. in image processing from Tel Aviv University. Most at home where the model, the data pipeline, and the physical system are all one job.

WORLD MODELS & ROBOTICS — HANDS-ON BUILDS

- **Action-conditioned world model on robot video (2026):** fine-tuned iVideoGPT — an autoregressive transformer world model — on the Bridge V2 manipulation dataset: **19.7 PSNR / 0.74 SSIM / FVD 446** on held-out action-conditioned rollouts (2 context frames, 14 predicted), vs 17.5 / 0.68 / 548 zero-shot from the OXE-pretrained baseline. Built the full pipeline myself — data preparation, training, evaluation — including diagnosing a silent Apple-Silicon numerical bug in the pretrained VQ decoder: token ids and transformer logits matched CPU exactly, but decoded pixels didn't, corrupting every rollout metric until isolated to the decode path; fix submitted upstream.
- **SO-101 manipulation policy in simulation (2026):** built the full LeRobot imitation-learning pipeline for an SO-101 arm in MuJoCo — scripted IK expert, a 600-episode / 93k-frame two-camera dataset at 50 Hz, and ACT (action-chunking transformer, 52M params) trained on Apple Silicon — reaching **88% pick-and-place success on unseen object layouts** (90% pooled over 100 trials), up from 68%. Warm-starting from the previous best checkpoint got there in 20k steps where training from scratch needed 60k, and checkpoints had to be selected on rollout success rather than training loss, which kept falling while success declined. Dataset keys and units match the real *so101_follower*, so the same code drives the physical STS3215 arm unchanged.
github.com/aviadarn/so101-lerobot
- **Performance work on an open-source world model (2026):** profiled Multiverse, Enigma Labs' open-source multiplayer world model, on Apple Silicon and traced its documented CPU-fallback requirement to one unimplemented MPS kernel under the pinned torch — bicubic upsampling was round-tripping to the CPU on every rendered frame — then found self-attention materializing a **1.1 GB** matrix per layer, twice per frame. Submitted PR: raises the version floor so the interpolation is untouched, and computes attention in head groups for **bit-identical output at half the peak memory**, with a reproducible inference benchmark.
[EnigmaLabsAI/multiverse#2](https://github.com/EnigmaLabsAI/multiverse#2)
- **Robot policy that fought back (2024, built with my kid):** autonomous JetBot on a Jetson Nano — hand-labeled ~3,000 frames, trained a behavior-cloning ResNet18 driving policy running in series with a YOLOv5 safety-stop detector on every frame inside a 4 GB / 10-watt envelope. First model trained on an artificially uniform dataset and failed on-road; three rounds of rebalancing the training distribution to match deployment conditions is what made it drive. Shortcut features and distribution mismatch stopped being abstractions after that.
- **Reproduced a published shortcut-feature result (2026):** independently reproduced the LeJEPa case from Enigma Labs' "Obsessed Encoder" research on rented A100s — three arms, 30k steps on ImageNet-1k, \$18 of GPU, measured before committing. Planting a faint *predictable* watermark drove the online linear probe from **14.5% to 0.51%** (chance 0.1%) while the training objective got **2.5x better** — and the matched-energy control,

differing only in whether the pattern repeats, landed back on the baseline at 13.1%, isolating predictability rather than added pixels as the cause. Paired-input cosines invert between the arms (0.888 vs 0.057): the encoder represents the key and discards the picture. Found and fixed two bugs breaking their documented quickstart along the way. github.com/aviadarn/obsessed-encoder-repro

EXPERIENCE

Nielsen Sports — AI Lead / ML Engineering Manager

2018–2025

- **Real-time video ML for a live public audience:** owned production computer-vision models on live sports broadcast — motion blur, occlusion, lighting shifts, camera cuts, unscripted chaos — served at 10K+ RPS across a 100+ GPU cluster where an outage is visible to viewers in real time.
- **Self-serve training platform:** built the Ray-based distributed training platform and data pipelines the team trained on, over datasets exceeding 100 million images; established "A/B against the incumbent before production" so every model shipped on measured evidence, not vibes.
- **Dataset and distribution ownership:** built dataset-creation and curation pipelines end to end — sampling strategy, labeling workflows, class balance — because in production video the training distribution decided more than the architecture did.
- **Inference performance engineering:** quantized models with TensorRT (FP16/INT8, dynamic batching via Triton) against strict latency and GPU-memory budgets; wrote a custom C++/CUDA inference server letting multiple models share one GPU memory arena — 30% more throughput, 20% lower latency.
- **Streaming video ingestion at scale:** event-driven pipelines on Kafka and KEDA over continuous multi-camera streams, GPU workers autoscaling against queue depth.
- **Led five engineers** while staying hands-on in production code — hiring, mentoring, design reviews, on-call.

A.Team — Applied AI Lead

2026–Present

- Build production AI systems for Fortune-500 clients; built a CLIP-based image–text evaluation pipeline grading generated visual content against a reference standard, with regression gates deciding what ships.
- Multi-agent LLM systems, RAG, and serverless pipelines on AWS — breadth beyond vision when the product needs it.

StuccoMedia — Software & Data Engineer

2014–2018

- Built distributed ETL pipelines, Spark clusters, Elasticsearch infrastructure, and cloud-native analytics platforms over high-volume event data.

TECHNICAL SKILLS

- **Models & training:** PyTorch, transformer world models (iVideoGPT fine-tuning on Bridge V2), imitation learning with LeRobot (ACT, behavior cloning), MuJoCo simulation, YOLOv5 detection, CLIP-based evaluation, Whisper fine-tuning, Ray distributed training on 100+ GPU clusters, dataset curation / sampling / distribution design
- **Inference & performance:** TensorRT, FP16/INT8 quantization, Triton Inference Server, CUDA, C++, custom GPU inference serving, latency/accuracy trade-off tuning, Jetson edge deployment, Apple-Silicon (MPS) ports
- **Data & infrastructure:** Kafka streaming video ingestion, KEDA autoscaling, Kubernetes, Docker, Argo Workflows, AWS, CI/CD, A/B release gating, high-volume data pipelines (Spark, Elasticsearch)
- **Languages:** Python and C++ in production, SQL, Bash

EDUCATION

M.Sc. Electrical Engineering (image processing) — Tel Aviv University, 2015–2018

B.Sc. Electrical Engineering (VLSI) — Shenkar College of Engineering, 2011–2015